

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459819.9226 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.016
Transit depth (Rp/Rs)^2	2.41 +/- 0.48 [%]
Orbital Inclination (inc)	85.86 +/- 1.35
Airmass coefficient 1 (a1)	1.155 +/- 0.017
Airmass coefficient 2 (a2)	-0.1074 +/- 0.0095
Scatter in the residuals of the lightcurve fit is	1.47 %
Best Comparison Star	None
Optimal Aperture	4.5
Optimal Annulus	10.17
Transit Duration (day)	0.0761 +/- 0.0081

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.155 ± 0.016 vs 0.1646 (deviation 0.56σ)
Duration fit vs published	0.0761 d vs 0.0758 d (deviation 0.03σ)
Mid-time O–C	-7.8 min
Depth SNR	9.0
Residual scatter	1.47 %

Light curve

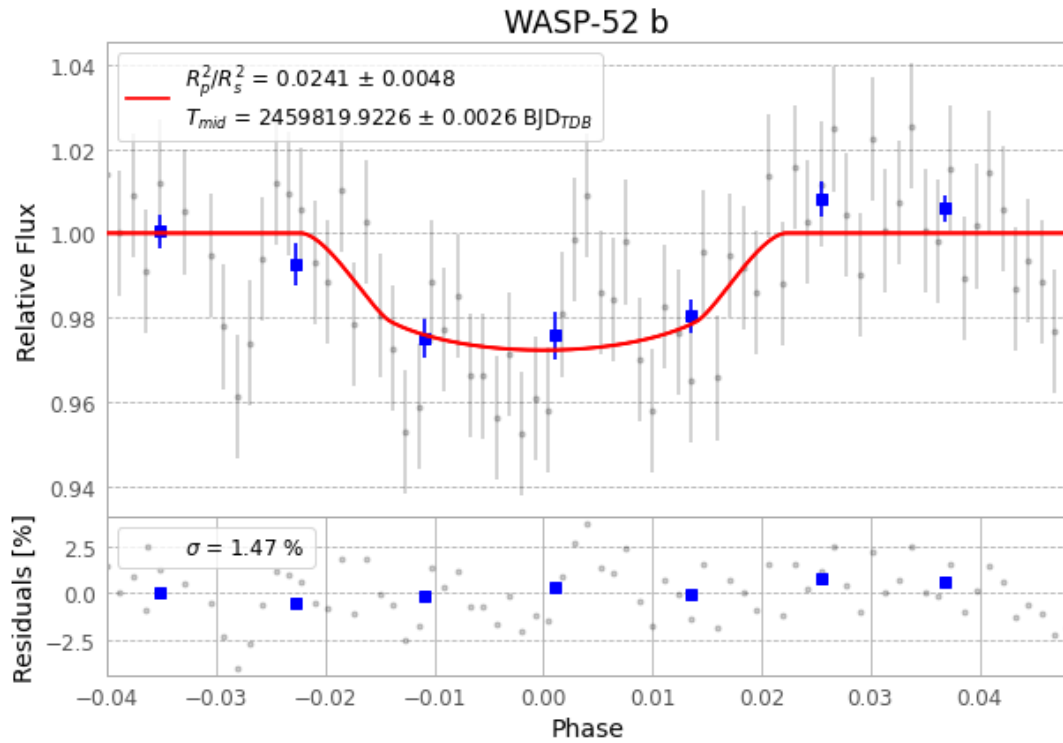


Figure 1 — FinalLightCurve_WASP-52 b_2022-08-28.png (SHA-256 cb59a4cb70d3f116...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [366, 138], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4034f6fb76bba0d9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

75 FITS frames (49 MB), WASP-52220828081813.FITS → WASP-52220828120012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-220828.log (6fa340c1a040...), FinalLightCurve_WASP-52 b_2022-08-28.png (cb59a4cb70d3...), FinalLightCurve_WASP-52 b_2022-08-28.csv (d6bcc200f6fd...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 21:49Z.